SED Hyperspectral → Excel Converter

User Guide & Installation Manual

Version 1.0

1. Introduction

The SED Hyperspectral → Excel Converter is a desktop application for batch-converting Spectral Evolution / NaturaSpec .SED field-spectrometer files into structured Excel workbooks. It is designed for agricultural field-trial workflows and remote-sensing ground-truth campaigns, where dozens or hundreds of point-spectra are collected per session and need to be assembled into a single tidy table for QA and downstream analysis.

Each conversion run produces two workbooks:

- **<name>.xlsx** - a spectral-only workbook containing ID, File_ID, and one column per integer wavelength (nm).
- **metadata_<name>.xlsx** - a full workbook including all SED header fields (date, time, GPS, instrument, integration, etc.) followed by the wavelength columns.

Both workbooks are styled with frozen ID columns, banded headers, and alternating row shading so they are immediately usable for review and analysis.

2. System Requirements

- **Operating system:** Windows 10 or later (one-click launcher); macOS or Linux are also supported via the Python script.
- **Python:** version 3.8 or later. The Windows launcher detects py, python3, or python on the system PATH.
- **Dependencies:** the openpyxl Python package. The Windows launcher installs this automatically into an isolated virtual environment.
- **Tkinter:** included with the standard Python installer on Windows and macOS. On Debian/Ubuntu Linux, install with: sudo apt install python3-tk.
- **Disk space:** approximately 30 MB for the virtual environment on first run.

Note: The launcher needs internet access only on its first run, to download openpyxl. Subsequent runs are fully offline.

3. Installation

3.1 Option A - Windows one-click launcher (recommended)

1. Download SED_to_Excel_Converter.bat from the project repository or the latest release.

2. Place it in any folder you have write access to. The first run creates a sed_excel_env\ virtual environment alongside the .bat file, so avoid read-only locations such as C:\Program Files.
3. Confirm Python 3.8+ is installed. If it is not, download it from https://www.python.org/downloads/ and tick **Add Python to PATH** during installation.
4. Double-click SED_to_Excel_Converter.bat. On the first run the launcher will:
    - Locate your Python installation.
    - Create an isolated virtual environment in sed_excel_env\.
    - Install openpyxl into that environment.
    - Launch the GUI.

Subsequent runs reuse the existing environment and start the GUI in a couple of seconds.

3.2 Option B - Run the Python script directly

Suitable for macOS, Linux, or any Windows user who prefers to manage their own environment.

```
# Clone the repository
git clone https://github.com/<your-username>/sed-to-excel-converter.git
cd sed-to-excel-converter

# (Optional but recommended) create a virtual environment
python -m venv .venv
# Windows:  .venv\Scripts\activate
# macOS/Linux:
source .venv/bin/activate

# Install the single dependency
pip install openpyxl

# Launch the GUI
python sed_to_excel.py
```

4. Using the Application

The interface is a single window with three input fields, a live preview of the output filenames, a Convert button, a progress bar, and a scrolling log.

Step 1 - Select the input folder

Click Browse next to Input Folder and select the folder containing your .SED files. Subfolders are searched automatically. The status bar reports how many .SED files were detected.

Step 2 - Select the output folder

Defaults to the input folder. Click Browse to choose a different location if you want to keep raw data and outputs separate.

Step 3 - Set the output file name

Defaults to hyperspectral_data. The two output workbooks are named:

- **<name>.xlsx** — spectral data

- **metadata_<name>.xlsx** - metadata + spectral data

The preview box updates live as you type.

Step 4 - Convert

Click ⚡ Convert SED Files to Excel. Progress is shown in the progress bar and live log. If output files already exist in the output folder you will be prompted before they are overwritten.

Step 5 - Review the results

A confirmation dialog reports how many files were processed and where they were saved. Files that could not be parsed (missing Data: block, malformed table, etc.) are listed in the log and skipped - the rest of the batch still completes.

5. Output File Structure

5.1 Spectral workbook — <name>.xlsx

Sheet name: Spectral Data. Columns:

ID	File_ID	350	351	…	2500
1	trial_2024_00001	0.0421	0.0438	…	0.1832
2	trial_2024_00002	0.0395	0.0412	…	0.1811

5.2 Full workbook — metadata_<name>.xlsx

Sheet name: Full Data. Same first two columns as the spectral workbook (ID, File_ID), followed by every SED header field that was present in at least one file in the batch, followed by all wavelength columns. Header fields are written in the order:

- Comment, Version, File_Name, Instrument, Detectors, Measurement
- Date, Time, Temperature_C, Battery_Voltage
- Averages, Integration, Dark_Mode, Foreoptic
- Radiometric_Calibration, Units, Wavelength_Range
- Latitude, Longitude, Altitude, GPS_Time, Satellites
- Calibrated_Reference_Correction_File, Channels

5.3 Identifier columns

Both workbooks include two identifier columns:

- **ID** - the trailing numeric portion of each filename (e.g. plot_00042.sed → 42), useful for quick numeric sorting and joining with other tabular data.
- **File_ID** - the full filename stem, preserved verbatim so each row can always be traced back to its source file.

The first two columns are styled with a blue accent fill and the pane is frozen at C2 so they remain visible while scrolling horizontally through wavelength columns.

6. How It Works

5. **Header parsing.** Each .SED file is read line by line. Lines before the Data: marker are interpreted as key: value pairs and matched against the controlled list of SED header fields above.
6. **Spectral parsing.** Lines after Data: are parsed as whitespace-separated wavelength value pairs. The first line after Data: (the column header) is skipped.
7. **Wavelength alignment.** All unique integer wavelengths across the input batch are collected and sorted; each row is filled column-by-column, leaving blanks where a wavelength is not present in a particular file.
8. **Excel writing.** The two workbooks are written with openpyxl, including styled headers, frozen panes (C2), 4-decimal-place number formatting on spectral cells, and 16-pixel row heights.
9. **Self-installing launcher.** On Windows, the .bat embeds the Python source as a base85+zlib-compressed string. On launch it locates Python, sets up a venv, installs openpyxl, decodes the embedded source to a temporary .py file, executes it, and deletes the temporary file when the GUI closes.

7. Troubleshooting

Issue	Resolution
“Python not found” when launching the .bat.	Install Python 3.8+ from python.org and tick Add Python to PATH during installation. Then re-run the launcher.
“Failed to install openpyxl.”	The launcher needs internet access for the first run only. Check your connection or proxy settings, then re-launch.
“No .SED files found.”	The tool searches recursively but only matches files with extension .sed or .SED. Confirm your files use that extension.
Some files are skipped with a parse-failure message.	Usually means the file is missing a Data: block or the spectral table is malformed. Open the file in a text editor to confirm. The remaining files in the batch are still processed.
GUI does not appear on Linux.	Install Tkinter: sudo apt install python3-tk (Debian / Ubuntu) or the equivalent for your distribution.
“Number stored as text” warning on metadata columns in Excel.	Expected — SED metadata values such as Date and GPS Time are stored as strings to preserve original formatting.
Output files exist; conversion is cancelled.	Click Yes when prompted to overwrite, or change the Output File Name to write to fresh files.

8. Credits

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